

Assembler, LED, and Virtual Pet Extension

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1 Assembler Implementation

Our assembler is composed of three main components: the emulate file, where instruction fetch and encode is performed; the modules, which consist of all instruction-encoding functions; and the data structures. The first data structure is the dynamic **symtable**, a dictionary of labels to addresses. The second data structure is the alias table, a dictionary of mnemonics to function pointers where the instruction type is handled.

This assembler utilizes two-pass assembly. The function **first_pass** loops through each line of the input assembly file, storing labels and their corresponding addresses in a symbol table. Next, the function **second_pass** accesses each line of the assembly file again, tokenising each line to identify the instruction name, replace labels with offsets, and extract instruction parameters. The instruction name is then inputted to the alias table's lookup function, which returns a function pointer to the proper encode function. For example, inputting the **add** instruction to lookup table returns a pointer to function **arith** in module **data-processing**. Each of these functions takes the list of tokens, the size of the token list, and an integer pointer to where the encoded instruction is to be stored. The integer pointer's contents are updated based on the instruction type and tokens. Each function also returns a status code indicating encoding success, either **EXIT_SUCCESS** or **EXIT_FAILURE**. Lastly, the encoded instruction is converted to binary and written to the specified output file. Note that in the second pass, any line consisting of only a label, directive, or newline is automatically skipped.

While writing our assembler, we learned to appreciate the importance of fleshing out useful data structures before implementing other functionality. For example, the dynamic symbol table was originally fixed-length, which is inefficient given there could be any number of labels in an assembly file. Also, the table initially stored addresses as integer pointers instead of just integers, which caused issues when looking up addresses using labels. Thus, thorough brainstorming, testing, and review are always necessary before a data structure is deployed. In addition, our group focused on the challenging implementation task we identified last week: writing utility functions early on. Throughout the assembler task, any function that seemed usable in multiple modules – like extracting immediate values and register numbers – were written in a file called **assembly-utils.c**. As a result, many of our tasks were simplified, and we enjoyed more efficient development.

2 Blinking LED Implementation

To implement Part III, we first built a circuit with one red LED, one 220Ω resistor, one wire to ground, and one wire GPIO pin 3. We then wrote our assembly code in a file called `led.blink.s`. Our program has five sections: `setout`, `sethigh`, `setlow`, `wait`, and an `.int` directives section.

In the first section, `setout`, we select pin 3 as the output by setting FSEL3 to binary 001 and storing this value in the given GPIO function selection address. The next section, `sethigh`, turns the LED on by storing 0x8 in the GPSET0 register address, which sets the bit corresponding to pin 3. Additionally, we set an “LED state” flag in register `w9` to signal that the current state is high before branching to the `wait` section of the code. Similarly, the `setlow` section turns the LED off by storing 0x8 in the GPCLR0 register address, setting the bit corresponding to pin 3 of the CLEAR register. This time, we set the “LED state” flag register to zero to signal that the current state is low before branching to `wait`.

In the `wait` section of our code, we use a loop to wait for a fixed amount of time before updating the output signal. First, we reset the loop by setting register `w3` to zero. Within the loop, we add one to register `w3` until it is greater than or equal to a set wait time of 0x50000. Once the wait time is reached, we check the “LED state” flag register to see if the current state is high or low, branching to the next state depending on the outcome of the comparison.

The final section of our code includes seven integer directives that are loaded into registers at the start of the program. The integer directives include the GPIO function select address, the GPSET0 address, the GPCLR0 address, the wait time, the GPIO configuration to select pin 3, the GP output configuration for sending high or low signals to pin 3, and the “LED state” flag.

One issue we faced while writing assembly was misunderstanding the instruction format. For example, our attempts to use the `mov` instruction to load an immediate value into a register yielded incorrect results, confusing us greatly. When we realised that our emulator and assembler do not support the functionality for immediate operands in `mov` instructions, we changed the code to use `ldr` instructions with integer directive operands to store the set and clear flags in the corresponding local GPIO output registers. Whenever a flag needed to be loaded into a register, we then used the local register as an operand to `mov` rather than immediate values.

Another challenge we faced was efficiently debugging our assembly code. Since our assembler does not support comments, descriptive integer directive names and section labels are essential to code clarity. However, our integer directive names were extremely generic in the beginning, causing confusion and debugging delays. As a result, we spent time adding clear labels for different sections and comprehensible names for integer directive, resulting in a more productive debugging process.

3 Virtual Pet Extension

A “virtual pet” is a digital toy that often requires care and attention. One example is Bandai’s Tamagotchi, a small toy with interaction buttons and a digital screen where the “pet” is displayed. Similar to a Tamagotchi, our virtual pet extension requires players to feed, clean, and play with their pet to keep the pet happy and alive. However, our extension also includes an “interaction limit” to prevent users from developing an addiction to their virtual pets. As a result, our extension can be a fun way to help children and teens practice responsibility and balance; while players must care for their virtual pet every so often, they cannot spend all day focusing only on their pet. This concept can then be generalized to any aspect of life, including studies, self-care, hobbies, and more.

3.1 Virtual Pet Design

Our extension includes three iterations of the virtual pet: one text-only version, one with a graphical display of the pet in a separate window, and one with the pet displayed on a screen in a Raspberry Pi. This creates three layers of accessibility: one for those who cannot load the GUI, one for those without a Raspberry Pi, and one for those who prefer a tiny, physical companion.

The virtual pet has three main categories that the user can interact with: cleanliness, happiness, and hunger. Each category decrements by one bar over a fixed amount of time, but can be improved by the user through keyboard inputs. If any category reaches zero, the current virtual pet cannot continue and the user would need to leave the game or restart with a new virtual pet.

We allowed the user to display their current category scores at any point, so they can check if any categories need particular attention. The program automatically warns the user if any category has only one heart remaining. Furthermore, we designed in-game coins that are regularly rewarded to the user if they keep their pet alive; these coins can be used to purchase gifts for and boost the happiness of their virtual pet. We later added levels as a way of encouraging user interaction; the longer a player plays for, the higher the level they can reach.

3.2 Challenges

Initially, we ran into some problems implementing the pet as a global structure pointer. Eventually, we realised the main issue was how the files were linked, which led to multiple declarations of the same variable. We fixed the linking issue by ensuring the virtual pet is only created once at the start of the program. This proved particularly helpful when we implemented in-game coins as a separate global variable, which involved some similar C syntax.

3.3 Testing

When we started designing our extension, we listed desirable functionalities we hoped to achieve. These included the user interacting with the virtual pet through keyboard inputs, decreasing the various category stats (such as happiness) at regular time intervals, and ensuring the number of hearts in each category is between zero and five. We referred back to this checklist to thoroughly test the program.

First, we confirmed that the stats would decrease at the expected regular intervals: every twelve seconds for cleanliness, eight seconds for happiness, and five seconds for hunger. To do so, we ran the game without interacting with the category being tested. We printed the stats at the expected time intervals to test if the stat for the current category was decrementing each time, and observed the successful outcome of the virtual pet dying when the category reached zero hearts. To test the user inputs, we waited for a particular category to have fewer than five hearts, then inputted the appropriate character. This test succeeds if we obtain an output message stating this category has increased by one heart as expected. For example, typing ‘f’ or ‘F’ for “feed” should print an output message confirming an increment in the hunger score.

We performed many more manual tests to determine whether the program worked as expected. In the future, we hope to test more rigorously by creating a test suite, as was used for the emulator and assembler, to test many possible states of our program and ensure it always operates in a predictable and desirable way. Due to time restrictions in this project, we decided it would be better to test the program in some of the most likely scenarios, and confirm if the main functionality

works rather than risk running out of time building a test suite and being unable to test anything at all.

4 Group Reflection

5 Individual Reflections

5.1 Tangina Kader

5.2 Amy Showell

I tried to communicate often with the others both online and in real life, which helped with group discussions and moving the project forward effectively. Through this project, I've realised that I have a tendency to rush and overlook details which can lead to code that is difficult to read or doesn't pass all the tests. Thankfully, Shriya and Yao pay more attention to detail than me, and noticed minor mistakes in my code which I missed; these could have easily become larger problems if they hadn't been caught until much later. In the future, I will slow down and focus more on the quality of my work as well as the quantity. Overall, I think I fit in well with this group. I engaged in group/pair programming and debugging several times, particularly for Part III. I clearly communicated the tasks I was working on and attempted to understand what everyone else was doing by asking questions and using other resources, such as Scientia.

5.3 Shriya Sistla

5.4 Yao Tong

Based on the Peer Assessment feedback, I fit into the group as an organizer who keeps the group on track through meetings, checklists, and deadlines. For example, I like to create Google Document task checklists for each section of the project, and a master checklist with key dates and deadlines. I also planned and scheduled a few group meetings where tasks are split and issues can be brought up. Planning meetings is not a strength I expected to have, and I am still developing my confidence in carrying out meetings. In addition to maintaining organization, another strength identified from the Peer Assessment feedback is writing and editing the content of reports for clarity. Unexpectedly, I also find it quite fun to structure the Latex document and edit for formatting. In terms of weaknesses, I often struggle with GitLab and forget to pull before pushing, sometimes leading to diverging branches that required more time to resolve. Since I have used GitHub before, I didn't expect this to be one of my weaknesses. Over the past three weeks, I have become exceedingly careful in checking that no new commits were made to the remote branch before I pushed my changes, and will continue to be vigilant in future group projects. Another thing I would do differently is to decide on and inform groupmates of reasonable "work hours". Currently, I think about the project during all hours, which is quite draining for me. Having a set time to relax will help with burnout and improve my wellbeing, allowing me to be more productive when I do work on the project. In the future, I hope to continue using my strengths to facilitate group organization through meetings, frequent updates, and checklists.